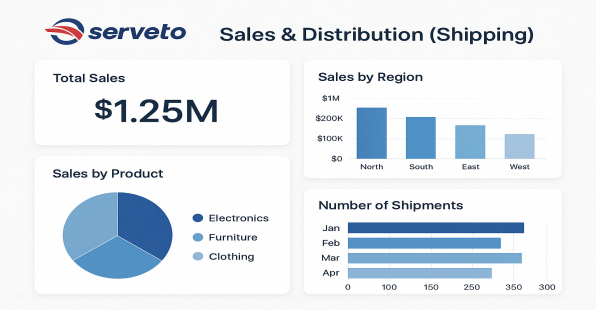
Serveto — Sales & Distribution (Shipping) dashboard

1) Overview & objectives

Serveto (third-party logistics partner) needs a dashboard for a premium customer that focuses on shipping and order KPIs. Source files: two order CSVs (two years) and a shipping Excel. The dashboard must include:



- Company logo and dashboard title under the company name.
- KPIs (for year 2021): Total Orders, Total Sales, Total Profit, Total Quantity, Average Discount, Profitability (Profit / Sales), and **Total Quantity for Standard Class shipping for the Accessories subcategory**.
- Selection controls (slicers) for Year, Region, State, Segment, Category, Subcategory.
- The KPIs must be interactive to slicer selections (but default to 2021).
- Five additional visuals must remain **static (not responsive)** to slicers/filters:
    1. Table: percentage of region-wide shipping by ship mode (no filters).
    2. Table: category sales by ship mode (no filters).
    3. Trendline: quantity by ship mode (no filters).
    4. Vertical bar: segment-wise regional profit distribution (no filters).
    5. Donut: regional distribution of orders (no filters).
- Only the KPIs should respond to slicers. The five visuals must stay intact (no interactions).

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2) Pre-load preparation & file naming

- Put the logo image in the project folder and name the file clearly (e.g., Serveto_logo.png). Avoid ambiguous names like log.png. Power BI inserts the image from file—clear name helps maintain version control.

- Place both CSVs and shipping Excel in the same project directory. Name the CSVs orders_2019_2020.csv and orders_2020_2021.csv (example). Clear names help automatic detection/merge later.

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3) Get Data (Power BI) & initial checks

1. **Get Data** → Text/CSV for the two order files. Import one at a time or use folder connector if they are identical-structure files (recommendation: use **Folder** to automatically combine identical CSVs).

2. **Get Data** → Excel for shipping file.

3. In Navigator, preview and choose **Transform Data** rather than Load — do cleaning in Power Query.

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4) Power Query transformations (recommended steps)

Treat both order files the same; combine into a single Orders table.

In Query Editor:

1. **Combine**

    - If using Folder connector: use *Combine Files* automatic step. If importing two CSVs individually: Append Queries → Append as New → select both.

2. **Header & types**

    - Ensure first row as header. Set data types explicitly: OrderDate → Date, Quantity → Whole Number, Sales / Profit → Decimal Number (or Currency), Discount → Decimal Number / Percentage, Postal Code → Text (if you need leading zeros).

3. **Regional settings**

   - If date parsing errors occur, use *Transform → Using Locale* to convert date columns correctly (select English (United States) if dates are MM/DD/YYYY).

4. **Trim & Clean**

   - Text columns: Trim and Clean to remove stray spaces/newlines (Product Name, Customer Name, Region, State, Category, Sub-Category).

5. **Remove duplicates & invalid rows**

   - Remove rows with missing OrderID or OrderDate (or fix if possible). Remove obvious duplicates (use all columns or subset).

6 . **Split columns if needed**

   - If ProductName contains codes, split or extract ProductID if helpful.

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7. **Create standardized fields**

   - Create Year, Month, MonthNumber, Quarter using Date functions for easier slicers.

8. **Shipping file**

   - Inspect shipping file. Typical fields: OrderID, ShipMode, ShipDate, ShipCost, Carrier, Region, TrackingID, ShippingQty maybe. Ensure OrderID type matches Orders'[OrderID] type (both Text).

9. **Join**

   - Merge Orders with Shipping by OrderID (Left Outer Join from Orders to Shipping if every order may not have shipping rows). Expand desired shipping fields. If shipping is at order-line level, ensure granularity aligns (e.g., one shipping row per order line).

10. **Surrogate keys**

- If needed create a surrogate OrderLineKey to uniquely identify lines (OrderID + ProductID).

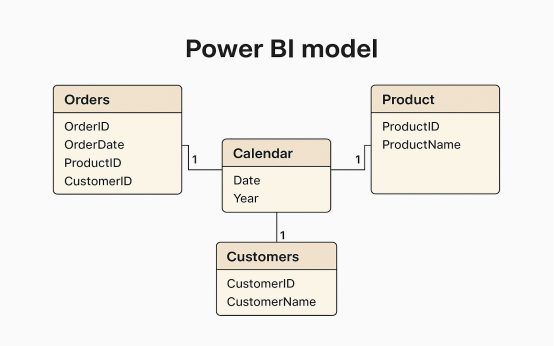
11. **Final checks**

- Remove unnecessary columns, reorder columns, and Close & Apply.

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5) Data modelling (Power BI model)

- Use a **star schema** pattern:
    - **FactOrders (combined orders + shipping)** — central table with OrderID, OrderDate, ProductID, Sales, Profit, Quantity, Discount, ShipMode, Region, State, Segment, Category, SubCategory, etc.
    - **DimDate** — build a proper date table (recommended). Create via CALENDARAUTO() or CALENDAR(MIN(Orders[OrderDate]), MAX(...)). Add Year, MonthName, Quarter, Fiscal Year if needed. Mark as Date table.
    - **DimProduct** (optional) — SKU/Product details.
    - **DimCustomer** (optional) — customer segmentation.
- Relationships:
    - FactOrders[OrderDate] → DimDate[Date] (many-to-one).
    - Other dims one-to-many to fact.
- Set default summarization for numeric fields where needed.



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6) Key DAX measures (copy & paste ready)

Use a Date table. Replace table/column names with your actual names.

Base measures

```
Total Orders = DISTINCTCOUNT( FactOrders[OrderID] )

Total Sales = SUM( FactOrders[Sales] )

Total Profit = SUM( FactOrders[Profit] )

Total Quantity = SUM( FactOrders[Quantity] )

Average Discount = AVERAGE( FactOrders[Discount] )  -- if Discount stored as decimal (0.1 = 10%)

Profitability =
DIVIDE( [Total Profit], [Total Sales], 0 )
```

Year-specific KPIs (if want measure scoped to a year or slicer)

If you use a Year slicer (connected to Date[Year]), these base measures will already respond to the Year slicer automatically.

If you want a measure forced to **2021**:

```
Total Sales 2021 =
CALCULATE( [Total Sales], 'DimDate'[Year] = 2021 )
```

Accessories Standard Class quantity (for 2021)

Option A — forced to 2021 (if requirement is strictly 2021):

```
Qty_StdClass_Accessories_2021 =
CALCULATE(
    [Total Quantity],
    FactOrders[ShipMode] = "Standard Class",
    FactOrders[Sub-Category] = "Accessories",
    'DimDate'[Year] = 2021
)
```

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Option B — responsive to Year slicer (if KPIs should use slicer selection):

```
Qty_StdClass_Accessories_SelectedYear =
CALCULATE(
    [Total Quantity],
    FactOrders[ShipMode] = "Standard Class",
    FactOrders[Sub-Category] = "Accessories"
    -- no explicit Year clause: Date filter from slicer will apply
)
```

Measures that must ignore slicers (for static visuals)

For the five visuals that must **not** change when users manipulate slicers, create measures using ALL() to remove external filters. Example: region-wide shipping percentages by ship mode.

Region-wide shipping count (no filters):

```
ShipCount_All =
CALCULATE( COUNTROWS( FactOrders ), ALL( FactOrders ) )
```

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ShipCount_ByRegionAndMode_All =

```
ShipCount_ByRegionAndMode_All =
CALCULATE(
    COUNTROWS( FactOrders ),
    ALL( FactOrders ),
    VALUES( FactOrders[Region] ),
    VALUES( FactOrders[ShipMode] )
)
```

PercentShipMode_Region_All =

```
PercentShipMode_Region_All =
DIVIDE(
    [ShipCount_ByRegionAndMode_All],
    CALCULATE( COUNTROWS( FactOrders ), ALL( FactOrders ), VALUES( FactOrders[Region])
),
    0
)
```

Use these measures in the static table visual; because they use ALL(FactOrders), slicer selections on Year/State/etc. will be ignored.

Category sales by ship mode (no filters):

```
Sales_Category_ShipMode_All =
CALCULATE(
    [Total Sales],
    ALL( FactOrders ),
    VALUES( FactOrders[Category] ),
    VALUES( FactOrders[ShipMode] )
)
```

Trendline quantity by ship mode (no filters) — use ALL(FactOrders) but keep time dimension:

```
Qty_ByShipMode_Date_All =
CALCULATE(
    SUM( FactOrders[Quantity] ),
    ALL( FactOrders ),
    VALUES( FactOrders[ShipMode] ),
    VALUES( DimDate[Date] )
)
```

Be careful: to get a proper trend you want DimDate[Date] on the x-axis and this measure on y.

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Regional distribution of orders (donut) ignoring slicers:

```
Orders_ByRegion_All =
CALCULATE( DISTINCTCOUNT( FactOrders[OrderID] ), ALL( FactOrders ), VALUES(
FactOrders[Region] ) )
```

Profit distribution bar (segment wise regional distribution):
Use a measure that calculates profit by Segment & Region but ignores slicers:

```
Profit_Segment_Region_All =
CALCULATE( [Total Profit], ALL( FactOrders ), VALUES( FactOrders[Segment] ), VALUES(
FactOrders[Region] ) )
```

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7) Visuals & layout (including logo + title)

- **Header area** (top row):
    - Left: Insert → Image → select Serveto_logo.png. Set **Fit** so the image keeps aspect ratio. Rename visual tooltip to "Serveto Logo".
    - Right next to the logo: Insert → Text box. Type dashboard title, e.g., **Serveto — Shipping & Distribution Analytics** (make font 20–28pt, bold). Align vertically with the logo. Use a subtle background or none.
- **Slicer pane**:
    - Add slicers for Year (single select or multi? requirement: selection option for year → use single-select dropdown to default to 2021), Region, State (hierarchy or separate slicer), Segment, Category, Subcategory. Place them in a left or top pane.
    - To default Year to 2021: pre-select 2021 in the Year slicer and save the report; consider creating a **Bookmark** named "Default View" and set as default on publish.
- **KPI cards**:
    - Place the KPI cards immediately under the header area — show the seven KPIs described. Use Card visuals or KPI visuals for current vs target comparisons if needed.
    - For the Accessories Standard Class KPI, label clearly: Qty (Standard Class — Accessories).
- **Static visuals area** (center/right/bottom):
    - Table 1: Region-wide shipping % by ship mode (use ShipMode and PercentShipMode_Region_All).
    - Table 2: Category sales by ship mode (use Category, ShipMode, Sales_Category_ShipMode_All).
    - Trendline: Date on X (DimDate[Date] or Month), measure Qty_ByShipMode_Date_All, legend = ShipMode.
    - Vertical Bar: Segment (x) by Region (stack) with Profit_Segment_Region_All.
    - Donut: Region with Orders_ByRegion_All.
- Maintain consistent color palette and axis labeling. Add titles/subtitles and footers (data as of date).

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8) Controlling interactions (critical)

You must ensure only KPIs respond to slicers; the five static visuals must ignore filters.

Approach A (recommended) — use Edit interactions:

1. Add slicers to the page and make sure they are connected.
2. For each slicer (Year, Region, State, Segment, Category, Subcategory):
    - Click the slicer → Format ribbon → **Edit interactions**.
    - For each of the five static visuals, click the **no-interaction** icon (circle with slash) so the slicer does not filter or highlight those visuals.
    - For KPI visuals, keep the default (filter or highlight) so they respond.
3. Repeat for each slicer. This preserves KPIs as interactive while static visuals remain unchanged.

Approach B — use DAX measures with ALL(...) to explicitly ignore filters (I provided examples). This is defensive: even if interaction is accidentally left on, visuals that use these measures won't reflect slicers.

Combine A + B for reliability: disable interactions and use ALL-based measures on static visuals.

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9) Slicer sync & default selection

- If the Year slicer is on multiple pages or you have multiple pages, use **View → Sync Slicers** to sync or isolate slicers.
- To set default selection = 2021, select 2021 on the Year slicer, create a **Bookmark** (capture data/visual state, not current page only), and set the bookmark as default/landing view. Alternatively, save the report with 2021 selected.

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10) Performance & optimization tips

- Use Remove Columns to drop unused fields to reduce model size.
- Convert text columns with many unique values into lookup dimension tables to reduce cardinality.
- Use SUMMARIZE/aggregations and pre-aggregate in Power Query for extremely large datasets.
- Avoid complex calculated columns that run row-by-row; prefer measures (DAX).
- Turn on **Auto Date/Time** off and use your own DimDate to reduce hidden tables.

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11) Validation & testing

- Validate totals: compare aggregated SUM(Sales) and COUNT(OrderID) between Power Query preview, model measures, and source CSVs.
- Edge cases: ensure orders with missing shipping rows are handled (e.g., exclude from ship-mode visuals or show as “Not Shipped”).
- Test slicer combination behavior: apply Year/Region/Category selections and verify KPIs update; verify the five static visuals do not change.
- Test quarter/month drilldowns, cross-filtering (KPIs should not filter other visuals by click; disable interactions as necessary).

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12) UX polish & accessibility

- Use concise titles and tooltips. Add hover tooltips to KPIs explaining calculation (e.g., “Total Orders = distinct OrderID in selected year”).
- Use consistent number formats: currency with 2 decimals, integers for counts, percentages 1 decimal.
- Add a small legend/annotation near static visuals noting “Static: not affected by slicers”.
- Accessibility: ensure color contrast and add keyboard navigation cues if enterprise users rely on them.

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13) Deployment & deliverables

- Save the .pbix file and source files in a versioned folder. Keep Serveto_logo.png included.
- Publish to Power BI Service workspace. For default view, apply the bookmark and set report to open to that bookmark (in service, set default bookmark).
- Provide a short user guide (1–2 pages) explaining: Where to change year (slicer), which visuals are static, and how to refresh the dataset.

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14) Example checklist before handoff

- Logo placed and titled correctly.
- Orders from both CSVs combined; shipping file merged properly.
- Date table created and marked as date table.
- All numeric columns are set to correct data types.
- KPIs implemented and tested for Year=2021 default and slicer interactions.
- Static visuals use ALL() measures or interactions disabled.
- Edit interactions set so only KPIs respond to slicers.
- Bookmarks created for default view (2021).

- Performance checks and data validation completed.
- User guide and source file attached.

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15) Common pitfalls & how to handle

- **Date parsing mismatches**: Use *Using Locale* transform and set the correct regional format.
- **ShipMode mismatches**: Standardize strings (e.g., “Standard Class” vs “Standard”) via Transform → Replace Values.
- **Counts vs lines**: Make sure Total Orders is DISTINCTCOUNT(OrderID) if you want unique orders rather than order lines.
- **Slicers incorrectly affecting static visuals**: Double-check Edit interactions for each slicer and visual; use ALL() in measures as backup.

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